

A High Performance Recommendation Framework For Online Learning Groups

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Abstract

For an online learner, they require a learning process and methodology that is flexible. The online learning system should be able to recommend suitable materials to a learner in a live environment. Much of the prevailing systems for online learning resource recommendation are working based upon collaborative filtering techniques. Such techniques have limitations when it comes to adapting to real-time and need the previous history of learners.

Hence, this study proposes to present a live learning resource recommendation methodology, appropriate for flexible and complex setups. The presented system is based upon reinforced learning challenges. The existing system explores the present environment to obtain data and exploit the information to make a decision. The proposed system will depend on live and changing data. The system being proposed is an enhanced recommendation method, which presents learning resource recommendation by obtaining student's behavioural information. Firstly, it will create user clusters based on their learning styles. Secondly, it enforces association rule mining to obtain all the choices and behavioural trends of each individual cluster. Finally, it will generate customised reference sets of differing sizes. The proposed system is a recommender methodology that references online materials for students towards their learning efforts depending upon the sameness of student's learning history and data mining approaches to uncover similarities between courses

Keywords: smart learning; expectation maximisation algorithm; e-Learning; apriori algorithm; online learning methodology; adaptive recommendation

1 Introduction:

Intelligent learning methodologies offer new avenues for imbibing knowledge and are expanding their user base and penetration over the recent years. Well-known online learning platforms such as Udemy, Coursera or Udacity are increasingly digitising learning materials for users with various educational backgrounds. The Web 2.0 technology has paved way for new important concepts for education online. Collaborative tagging, clubbed as part of Web 2.0, is considered a popularly used tool for classifying live online content. Spectacular user experiences together with collective intelligence impact the selection of command on educational platforms. However, short of ample guidance, users or learners can come into face with difficulties in selecting appropriate learning resources due to the huge information overload in the course of their learning. Observing all trends in the current world stage, the process of combining collaborative tagging methods and recommender systems into an online learning system is a deliberate

one. The process of collaborative tagging sets up a set of tags, useful to describe a resource. The process of tagging allows the user to use their own words or notions of greater importance. The system being proposed applies an algorithm of clustering upon the course-dataset. This is needed for grouping similar learners into clusters based on their previous online behaviour. The Expectation Maximisation clustering algorithm, together with the Euclidean distance is employed to quantify those similarities among courses. Learning methodologies are particular traits which learners start to focus on, develop, imbibe, and store new and often difficult new information. Obtaining understandings into the different learning methodologies offer ways to propose and present recommendations that are customised to each individual's demands. Choosing an appropriate model is the deciding factor in integrating learning methodologies into evolving learning environments.

The paper has been organized as follows: Section II details existing research works relating to the topic. Section III introduces the online learning style methodology. Section IV presents the recommendation system based upon online learning methodologies and the different styles. The framework architecture and its modules, is explained in Section V. The evaluation of the framework has been discussed in Section VI, result set and observations make up Section VII and Section VIII draws up the conclusions for this paper.

2 Related Work

This particular section describes all the various current e-learning solutions that have been discussed.

Learning styles are detected more accurately by the use of techniques such as VARK learning style (Visual, Audio, Read and Kinaesthetic) recognition, Hybrid model (recognition based on literature and automatic recognition) [1]. However, this technique uses questionnaires for recognising learning styles. The prevalence of questionnaires acts as an obstacle due to which the detection results come out less precise.

Aleksandra [2] proposed a system which puts its onus on appropriate selection of collaborative tagging methodologies which would result in enhanced and personalized recommendations, which will be in alignment with the learner's interests, their learning style, demographic traits and previously imbibed knowledge. But this requires extensive peer interest and tagging for rare learning resources to be recommended to beginner or intermediate learners.

Pereira [3] employed a method which extracts data from the Social Networks using Virtual Learning Environment (VLE) analysis allowing the identification of the different information about their profile, interests and behaviour. However, access to social media data requires user and administrator permission. The evolving privacy laws make it difficult to acquire data. Fake data is also a deterrent to a robust recommendation system.

Using the help of big data analytics, the system expansively records, tracks and master the various learning traits, learning demands, learning base and the behaviour of learners while learning, and for various types of learners [4]. SOALS-Service Oriented Adaptive Learning System and Network Collaborative Filtering Recommendation Technology is used to achieve this. Since the system does not provide any satisfactory performance in the academic performance of the outgoing students, the current generation is sceptical about it.

The adaptive technique used in this system assists in solving the issue of low rating obtained despite the suggested matched items and the high score for the non-matched materials employing the Learning Style based Information Seeking Tool (LSIST) and Adaptive algorithm (kinaesthetic and aural) [5]. The

Reading Material utilised is in a format of PDF. This then restricts the ability of the adaptive method to satisfy audio-preferring students completely. Reading material in audio format is not available to the LSIST. This is a huge limitation for aural students.

Mejia [6] has presented an inclusive user model for dynamic recommendation to assist learners with dyslexia or reading challenges in virtual learning. However, it is unable to give a clear picture of learners and all of their differences and their satisfaction with the recommendations.

The system automatically detects learning styles using SVM and Data Structures [7]. SVM has an overview performance that can prevent the choosing of local minima structure. It does not depend on human behaviour and knowledge. It is ideal for solving highly-dimensional cases. Pre-processing the information using textual-based techniques is time-consuming & inefficient.

Thanyaphongphat [8] proposed a system which can encourage positive outlooks in an adaptive online-based learning environment using Adaptive Learning, Kolb Learning Style and Felder-Silverman Learning Style models, learning dataset combined with SQL. However, it has difficulty in understanding achievements, delivering content, its effectiveness, and the acceptance of those.

The system solves present learning method detection challenges. Problems of detection method with present normal and automatic learning can be progressed using Learning style, Prior Knowledge, Prediction, Weight Cosine Coefficient (WCC) algorithm [9]. Learning style is evolving, depending on the issue that someone is currently learning with reliability and other problems. The existing style has a weakness in the neighbourhood of time, reduced effectiveness and increased subjectivity.

Sfenrianto [10] has proposed a system which uses Triple-factor (learning methods, motivation and information grabbing ability) to better understand the presence of these parameters to assist a model of value creation in e-Learning process. However, the onus for calculating the functioning of e-Learning is not done mostly.

EDM - Educational Data Mining and Apriori algorithm are used to construct models through analytics and mining of learning-related data [11]. Diagnostic techniques can be based upon statistical styles, techniques based on distance or proximity, and techniques based upon the densities. No specific technology is used to protect the data that is retrieved and hence there are high chances of data being misused.

Shan Liu [12] proposes the hybrid collaborative categorising algorithm constructed upon news recommendation which is employed to predict users' rating scores to create a user-rating matrix with non-zero values. It increases the accuracy and stability by involving the hot factor to bring down the relative significance of popular materials to find similar users. However, it requires background knowledge on collective filtering algorithms and is susceptible to errors in real world applications since the experiment has been conducted on a closed set of data.

The system deploys a hybrid technique that makes use of both Content and Collaborative filtering algorithm into consideration for practical recommendations [13]. Produces low Mean Absolute Error (MAE) and sparsity for small datasets. Results vary when applied to real world large datasets.

Furukawa [14] proposed a system to recommend teaching materials by visualizing free imageries in the courses, which helps to clarify the differences between learners. This will be a boon to recommend supporting learning items suitable for the users. Uses conventional method of pre-processing data, i.e. learners have to provide some input for proposed system to work, which is time consuming and also provides not so accurate results.

Able to adapt teaching and learning methods via social interactions and technology [15]. Through LMS, the analysis produce data that assists administration in making rules and policies, to survey and measure users' performance and grade them more precisely that will benefit users and the accompanying learning process. Users differ from one another in various ways and the use of LMS, the static learning behaviour dataset has a high chance of giving inaccurate results for individuals.

Moubayed [16] proposes that with machine learning and data analytics, the domain of e-learning is examined in terms of definitions and traits. Requires background knowledge of machine learning and data analysis techniques for personalisation of user content.

The system uses a completely unique group-oriented recommendation algorithm supported learning generative networks, which are produced by charting out the learners' learning log to a private learning generative network (PLGN) based upon an information map [17]. Involves processing through a huge amount of data to produce the required logs and subgraph recommendations.

Rahman [18] proposes a system which is able to categorise Web search engines with customised recommendations of search results which adjusts to users' learning abilities and behaviours. The existing system of the proposed paradigm has a limitation because the customised Google search API restricts it to just over 100 free search queries each day.

This recommender algorithm employed with the system helps people in receiving personalized suggestions for learning according to their learning styles [19]. Makes use of Felder Silverman Learning Styles Model (FSLSM), Mean Absolute Error (MAE), Root Mean Square Error (RMSE) measures and Hybrid Filtering (HF). This system was not tested with large number of students resulting in the chances of getting false recommendations for students of different learning capability.

All of these existing systems aim to recommend learning materials very accurately, but most of them use group-based collaborative filtering and does pre-processing of information employing scripture-based techniques, which is time-consuming and ineffective. The algorithm being used, K-means, fails to produce clear and correct results when the clusters are circular. Our enhanced approach is aimed at overcoming those disadvantages and proposes a live recommendation technique which has a low computation cost, best suited for flexible environments, depending not on human behaviour and knowledge and is enabled to detect learning methods more precisely.

3 Online Learning Style Methodology

The conventional and modern learning styles vary starkly in various ways. This work compares both these styles on a set of parameters: motivation, communication, visual and auditory preferences, sensory and sequential choices and global preference. These preferences can be explained as follows – students learning in a traditional style might have preferences on the temperature, light and sound in the classroom; whereas, in an online platform these choices are irrelevant as the main element of interaction for the student is the course webpage. The parameters can be categorised as Emotion (Motivation), Physiology (Visual and Auditory), Sociology (Communication) and Psychology (Sensory, sequential and global preferences).

(1) Motivation. Borrowing the thought principle from David Entwistle's model for psychology, motivated learners on the online platform are self-directed. This means that they are more likely to monitor the course catalogue and click on unpopular learning resources. This means that they are more open to knowledge reception by using e-learning materials and will simply finish the required

assessments and materials. In contrast, the non-active or unmotivated learners are likely to drop out or slow-down in their course progress.

(2) **Visual.** Learners online vary in how much they prefer videos and pictures in their courses. They receive information from pictures, diagrams, symbols and sights. Users preferring visual media are seen to understand content better easily.

(3) **Auditory.** Those learners who prefer to take full or part of their courses supported by verbal or audio support materials have improved performance when it comes to the speed at which they comprehend topics. Although the depth of understanding is less than visual media, auditory media makes the learner think more. The Felder-Silverman model is used to describe the visual and auditory parameters. This model allows students to select their individual learning style and teachers to offer courses in diverse styles to suite students with various learning styles.

(4) **Sensory.** Learners can be segmented on the basis of what they prefer out of data, facts and research (experimental) materials.

(5) **Sequential.** This parameter defines learners who prefer to browse and take their course in a logically arranged order or progression. The order of the course maybe set by the course instructor or ordered on the basis of peer course taker review.

(6) **Global.** Learners can be classified by the order in which they take a learning resource after another. Global learners refer materials from different knowledge sources and does not stick to the course webpage alone. Active learners are more likely to be a global learner.

All these above-mentioned parameters can be classified into three classes:

1. Emotion – dealing with the feelings of the human mind.
2. Physiology – the bodily preferences of a human subject.
3. Psychology – the psychological preferences of the human subject.

In the traditional learning model, learners could not be both sequential and global learners. This is because both these learning were mutually exclusive. But in this proposed online system, the learner can be both sequential and global. This is possible because if a learner receives high scores in these styles, it means that the student is flexible in their learning style from various online sources.

Feature	Class	Description
Motivation	Emotion	Preferring assignments, lecture duration, etc.
Visual	Physiology	Preference of videos and pictures in lectures.
Auditory		Preference of auditory and text materials.
Sensory	Psychology	Preference of facts, experimental materials, data, etc.
Sequential		Learn course arranged logically in an order.
Global		Skipping to another material from one.

The proposed system presents the comprehensive features of the online learning system and the online learner. It also explains how to differ it from the traditional systems. The features are selected and measured by means of the learner browsing history and their history on learning platforms. Grouping of these features into classes are done based on the clinical description of these parameters.

4 Recommendation System Based On Online Learning Style Model

1. User Clustering

Clustering is amongst the most common exploratory data analytics technique for getting an idea of the structure of the data. It can be used to identify the subgroups in data which have the common data points within the subgroup but the data points outside the subgroup are different. This study uses the Expectation Maximisation algorithm, which works similar to K-means but uses probabilities instead of means to form clusters. It quantitatively gives an output of the probability that an item can be the member of a cluster. The similarity of the data points are quantified using Euclidean-based distance.

Taking a set of N learners, $P = \{p_1, p_2, p_3 \dots\}$, where each individual learner p_N is replaced by a vector $\{d_{N1}, d_{N2}, \dots, d_{N6}\}$ based on the online learning style methodology, where d_{Ni} represents the i^{th} feature or parameter of the individual learner.

$$d(p_a, p_b) = \sqrt{\sum_{i=1}^6 (d_{ai} - d_{bi})^2}$$

After obtaining the Euclidean distance, the Expectation Maximisation (EM) algorithm can segment the N learners into K disjoint set of clusters $C = \{c_1, c_2, \dots, c_k\}$. The concept behind the Expectation-Maximisations algorithm is that after several iterations, some criteria for a convergence is met.

The EM clustering algorithm is as follows:

Input: A pre-defined set of clusters k, stopping criteria and a dataset.

Output: A set of weighted k-clusters.

Expectation stage: Per each dataset record p, calculate the probability of membership of x in the clusters $y = 1, \dots, k$.

Maximisation stage: Update parameters of the model with the probability weights.

Termination criteria: If the termination rule is satisfied halt, else set $e = e + 1$ and move to (i).

The Expectation-Maximisation algorithm trumps over K-means algorithm where there are circular clusters involved.

Since K-means is based on calculating means and so if two concentric clusters with different radii are present, K-means will fail to classify items to both the clusters since the means of both the clusters will be too close to be differentiated.

The EM succeeds here by calculating probabilities and thereby able to differentiate the concentric clusters.

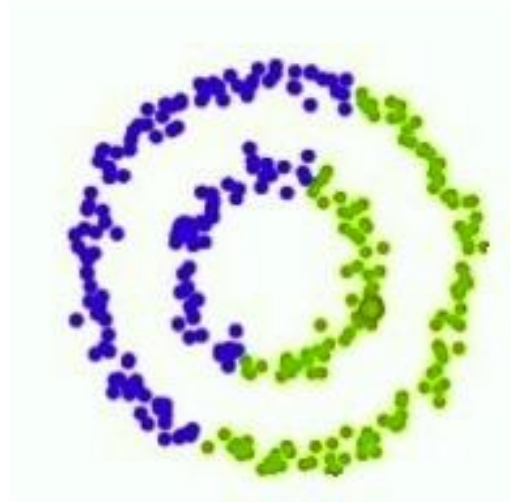


Figure 1 Concentric Circular Clusters

2. Extracting User Preferences

In order to mine useful learning preferences from the user's browsing data, association rules and calculation of similarities is done for each of the learners' cluster c_k . The association rule searches for items that are repeatedly present in the dataset and attaches it to a larger common set.

Provided a set of M learning resources $L = \{j_1, j_2 \dots j_m\}$, the system uses the cosine similarity matrix to determine the similarity $\text{sim}(j_m, j_n | c_k)$ between the resources j_m and j_n , based on the browsing history of the learners in the cluster c_k ,

$$\text{sim}(j_m, j_n | c_k) = \frac{|S(j_m | c_k) \cap S(j_n | c_k)|}{\sqrt{|S(j_m | c_k)| |S(j_n | c_k)|}}$$

where $|S(j | c_k)|$ symbolises the quantity of users in a particular cluster c_k that clicked on a particular material (item) j and $|S(j_m | c_k) \cap S(j_n | c_k)|$ is the number of learners in cluster c_k that selected resources j_m as well as j_n .

Association rule being an underlying statement of the type $A_i \rightarrow B_j$, which implies that the presence of A_i defines the presence of B_j .

Confidence is the dependability of a rule that also signals to the importance of an association among the materials. This proposed study implements the Apriori algorithm to produce association definitions which satisfy required confidence from recurrent item sets, where its support surpasses a learner-specified support threshold.

The confidence for association rules, for instance $A_i \rightarrow B_j$, which shows the group of learners who selected both resources i and j can be calculated as,

$$\text{conf}(A_i \rightarrow B_j | c_k) = \frac{\text{support}(A_i \rightarrow B_j | c_k)}{\text{support}(A_i | c_k)}$$

3. Learner's Resource Recommendation

Based on the results of resource similarity, the system makes predictions through calculating the interest for a user x in a cluster c_k for an item j_m as,

$$P(x, j_m | c_k) = \sum_{j_n \in S(x) \cap S(j_m, L)} \text{sim}(j_m, j_n | c_k)$$

where $S(x)$ being the set x viewed and $S(j_m, K)$ representing the highest of the likeliest learning material to j_m .

The application of the recommendation system and the clustering algorithm on the reduced database clusters brings down the computational demand and focusses on identifying trends and observations from users with likewise learning methods. Employing association rules having higher confidence metrics to filter the output of item-based filtering, the enhanced methodology concentrates on personal choices and an item-item relationship resulting from the clusters. The enhancements in accuracy recommends materials without pegging the measure of the recommendation set.

The tendency of a user x from a cluster c_k can be defined as follows:

$$T(x, j_m | c_k) = P(x, j_m | c_k) \times \sum_{I_j \in F(x)} \text{conf}(I_j \rightarrow \{j_m\} | c_k)$$

where $P(x, j_m | c_k)$ is the interest of a learner x of a cluster k on a particular resource j_m and $F(x)$ being the set of materials recurrently compiled by the browsing history of the user x .

For Data Visualization purposes, the system makes use of Matplotlib, which is a 2-D plotting library available in Python that helps in emulating Matlab like graphs and visualizations. Matplotlib has a sub-library, matplotlib.pyplot which aids in plotting the graphs and figures. Pandas, a data analysis package is used to import the required datasets for plotting through the command 'pd.read_csv()'. This results in the generation of 'Pandas Dataframes', which implicitly uses matplotlib.pyplot to generate the graphs. One of the many advantages of using this library is the fact that it can plot many sets of data together.

5 Framework Architecture

1. Dataset Processing

The dataset is obtained in a CSV file format. The Python API along with Pandas, a powerful data analysis package, is used to load the CSV files. Once loaded, a data frame is returned which is used for summarizing and plotting.

The header in a CSV file is used in automatically assigning names or labels to each column of the dataset. If the file does not have a header, the attributes will have to be named manually.

The comments in a CSV file can be identified when a line starts with a hash sign (#).

Depending on the method chosen to load the machine learning data, it has to be determined if these comments are to show up, and how it can be identified.

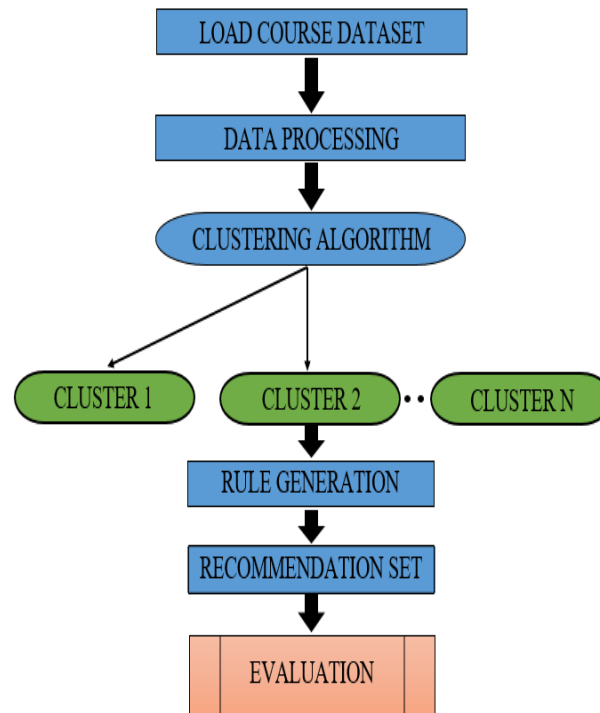


Fig 2 Architecture diagram of the system

A delimiter separates multiple values in a field and is indicated by the comma (.). The tab (\t) is another delimiter that can be used, but it has to be specified clearly.

If field values in a file contain spaces, these values are often quoted and the symbol that denotes this is double quotation marks. Other characters may be used, but it has to be specified clearly in the file.

2. Investigative Data Analysis

Investigative Data Analysis (IDA) is an initiative included within the data analytics process.

Data analysts and research faculty use IDA to gain insights and recapitulate the innards of a dataset, stereotypically with a particular question prepared, or rather for preparing towards an innovative statistical modelling in later stages of data analytics.

IDA depends on data visualizations which allow researchers to recognise and define trends within the database, which otherwise would not have been visible.

This study proposes to use the Expectation-Maximisation algorithm for clustering the sample of users by identifying the common data points by calculating the probability that a data point can be the member of a particular cluster. It scales well with large data sets by being able to scan cluster sets and provides reliable and precise results. Using Euclidean distance to calculate the similarity between two samples, the sample set is divided into more disjoint sub-cluster set.

Investigative data analysis is employed to authorise technical and resource conventions, and to recognise trends. The assumptions which analytics teams tend to make on raw databases can be categorised into one of the two categories - technical conventions and resource assumptions.

To maintain assurance that supreme analytical algorithms and models are used in data analytics, and that the resulting are deemed to be precise, particular assumptions on the technical side on the data should be accurate.

3. Data Analysis and Recommendation

The concept of association rules are used to determine the learner preferences on the platform. It is normally used to uncover relations between different variables in huge databases.

Cosine similarity metric is used to quantify the similarity between the learning resources based upon the browsing history of the learner. Depending on what the user might have read or studied previously, relevant new materials can be recommended.

The Apriori algorithm is used for the generation of association rules. It generates rules that satisfy a minimum level of confidence metric. It is illustration of how much of which content users have been clicking on.

Recommendation set is the collection of resources which is compiled by the system based on the association rules and the learner history. Based upon the results of similarity of the resources, the system makes predictions of what the learner might want to take next.

The prediction is done using the interest metric. It calculates the interest that a learner who belongs to a cluster, can have on a resource item.

The interest metric uses the cosine similarity metric to analyse the resource in question by comparing it with the set of resources viewed or studied previously by the learner.

The recommendation system, after cleansing the data, makes use of the reduced dataset to lower the computation cost and focusses to find the patterns that can exist in the resource dataset.

Evaluation is the final phase when the system is deployed on the data and the observations are prepared based on the output.

6 Evaluation Of Learning Model

The system uses the Open University Learning Analytics Dataset (OULAD) which contains anonymised data about courses, students and their related interaction with Virtual Learning Environments (VLE), segmented into several tables for seven different modules connected by unique identifiers.

It contains data of over 32,500 learners, their assignments and assessments and the logs of their exchanges with the VLE. In this particular presentation, the onus is on the information from the learning environment (VLE), which describe the learner preference of the resource materials.

The evaluation part consists of data preparation, user clustering and analysis or recommendation.

1. Data Preparation

The process of data preparation consists of the subsequent steps:

a) **Data Cleaning:** Valuable and relevant data is selected by filtering out learners whose average assignment score is higher than 60 on a scale of 1 to 100. Learners with a score of less than 60 is not considered to be a regular learner or a reliable source of data.

b) **Aggregation of data:** To populate the data aggregates, the number of clicks by each user on different types of resources are analysed. The outcome is a vector table which lists the user horizontally and shows the preference for each resource grouped by users.

For example, the number of times a learner clicks on a audio module in a lesson can be used to analyse the auditory parameter preference for that particular user.

c) **Removal of outlier variables:** In order to normalise the results of the system, since it relies on the metric for Expectation-Maximisation clustering which is Euclidean distance, which can output abnormally biased numbers. Features with more standard deviation than 10 is removed.

d) **Selection of feature:** This is the most important step where using data mining, the most relevant parameter or feature is selected.

2. User Clustering

The learners are divided into clusters based on the features. The most important step is selecting the number of clusters (k) of the clustering process. The Calinski-Harabaz (CH) index is used to evaluate the feasibility of a model. A high CH index is the indicator for a well-defined cluster, whereby the clusters are aptly differentiated.

In the diagram (fig 3), the CH index spikes when the number of clusters is set between 5 and 6. Based on the parameters chosen, the number of clusters (k) is set at 6. This gives the highest CH score, meaning the best separated and meaningful clusters for the data.

The Expectation-Maximisation (EM) clustering algorithm calculates the cluster sum that is weighted using probabilities. It calculates the probability that a resource item can be the member of a particular cluster.

In this study, some learning resources fall into one or more clusters simultaneously. This is because it can be the part of multiple topics.

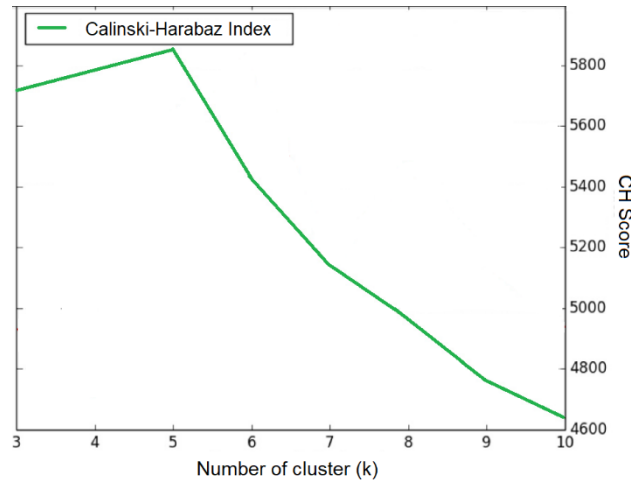


Fig 3 Plot of CH score against the number of clusters

The EM algorithm supports mixed membership and so the resource is defined as belonging x-percent to cluster 1 and y-percent to cluster 2.

3. Analysis/Recommendation

The two most important metrics used for recommendation is recall, which is repetitive taking of a resource and precision, which is the relation of the material to what the learner is seeking.

However, both recall and precision cannot be enhanced simultaneously and is incompatible.

For instance, recommending a later chapter to a learner for recollection when is user has reached only the preceding chapter is unproductive.

The prevailing system, employing the time consuming K-means clustering algorithm, even though with a 94.746% successful classification instance, fails when it comes to the speed of recommendation. It has a high overhead of computation cost and time and is not able to perform classification for huge multi-dimensional datasets on a real-time basis.

Finally, the proposed system, employing the Expectation-Maximisation (EM) algorithm with 87.429% fitted successful classification instance, performs superior in terms of the computation time and the adaptability to a real world online learning platform. It also fares prominently better when the clusters involved are of abnormal shapes and is able to produce precise results. With mixed memberships supported by the EM algorithm, items can be the part of multiple clusters simultaneously although in different proportions. This enables users to be partly segmented into multiple clusters, increasing the variety and subjectivity of the clusters or groups. The personalisation aspect of the system is hence enhanced.

7 Result Set And Observations

The comparison of K-means and Expectation-Maximization (EM) clustering algorithms can be done with the help of an artificial dataset such as OULAD. The algorithms are first applied to the dataset to get the appropriate clusters and results, after which comparison is done.

From Fig 4, it can be seen how EM is clearly effective in categorising data, or in this case materials and users in different percentages to different clusters. This is possible since EM uses the concept of mixed probabilities, i.e. the probability of an item partly belonging to one cluster and partly belonging to another cluster. This allows for the learning resources to be allotted to more diverse types of learners.

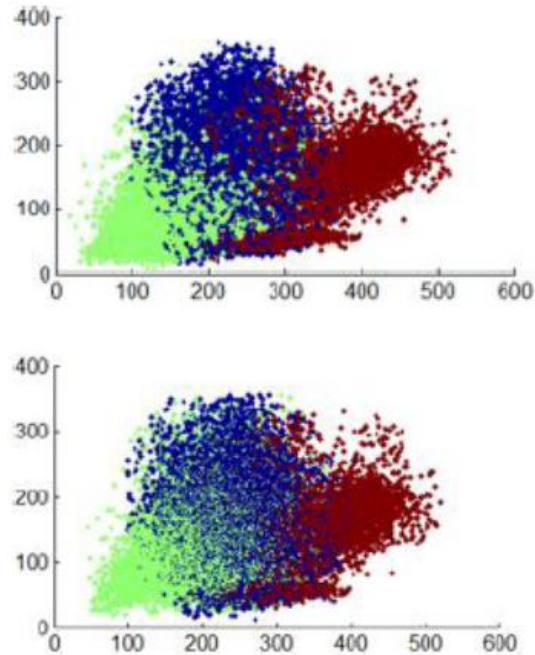


Fig 4 K-Means Clustering (Top) Vs EM Clustering (Bottom)

K-means on the other hand uses the concept of cluster centre mean values to do the clustering. Spherical clusters with similar radii, have the cluster centres close by and K-means confuses the cluster centres and gives inaccurate results. Since EM uses the mixed probabilities concept, it has a better scope at clearly categorising spherical covariance or spherical cluster data than K-means.

Fig 5 illustrates the cluster analysis of K-means and EM on the OULAD dataset. On the top-left is the original data, which has some outliers marked as yellow points. Pre-processing of data is done to alleviate the drawbacks of outliers on clusters. In the top-right is the cluster which is obtained after the application of k-means, which shows that the data is split. Finally, on the bottom is the EM cluster, which shows that it can point out the normal distributions exactly. This is possible since EM is not restricted to spherical clusters, but can be modified to obtain different shapes of covariance matrices such as diagonal and generic.

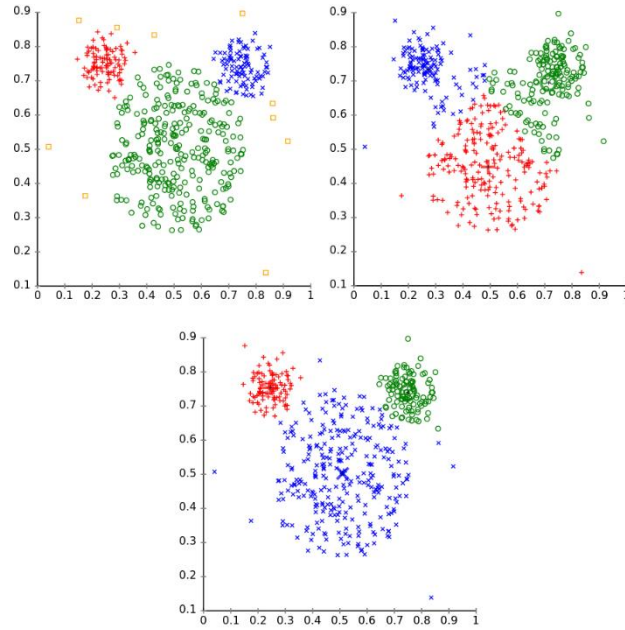


Fig 5 Cluster Analysis

Table 1 represents the various attributes belonging to the experimental OULAD dataset, which will be used to analyse and contrast the performance of K-means and EM algorithms. The experimental data for performance comparison are bound by attributes which is approximately around a total of 13 samples. Each of the attributes are well defined in Table 1.

No.	Attribute	Type	Range
1	Gender	Binomial	0: Male, 1: Female
2	Previous Attempts	Numeric	[3.80, 15.90]
3	Studied Credits	Numeric	[0.08, 1.58]
4	Assignment Submission	Numeric	[0.00, 1.66]
5	Score	Numeric	[0.60, 65.80]
6	Student Count	Numeric	[0.01, 0.61]
7	Course	Numeric	[1.0, 289.00]
8	Course Module	Numeric	[6.00, 444.00]
9	Assessment Engagement	Numeric	[0.99, 1.04]
10	Peer Participation	Numeric	[2.72, 4.01]
11	Audio-video Content	Numeric	[0.22, 2.00]
12	Course Duration	Numeric	[8.00, 14.90]
13	Student Performance	Nomial	[0] very bad, [10] excellent

Table 1: Attributes of experimental OULAD data

An experimental result is obtained by applying the K-means clustering algorithm on the experimental OULAD dataset.

It can be seen that from Table 2, K-means has a 94.746% successful classification instance.

Instance Classification	Percentage
Correctly Classified	94.746 %
Incorrectly Classified	5.253 %

Table 2 Classification result applying K-Means

Another experimental result is obtained by applying the EM clustering algorithm on the experimental OULAD dataset. Each of the 13 sample values belonging to each attribute were clustered. It can be seen that from Table 3, EM has an 87.429% fitted successful classification instance.

Instance Classification	Percentage
Correctly Classified	87.429 %
Incorrectly Classified	12.570 %

Table 3 Classification result applying EM

The accuracy of K-means (94.746%) is evidentially higher than EM (87.429%), but at the cost of slower processing speed and higher processing time. This results in an overall slower recommendation system. EM on the other hand, even though with a lower accuracy percentage, has a quicker processing speed and is able to perform better in clusters where the covariance matrix is of abnormal shapes. The clusters obtained in the proposed system from the OULAD dataset are mostly of spherical/circular shape and have centres very close by, so K-means does not perform well. Also, the better speed of EM will make it more suited to perform in the live environment with real time data. It'll offer a fluid experience.

8 Conclusion And Future Work

Over the past few years, recommendation-based systems have had a lot of success and improvements, particularly in the field of online learning due to the age of information explosion and the many data analytical and exploitation tools and studies that came along with it. Diverse advances, such as collaborative filtering, hybrid data sets and other methods were combined with many of the existing smart learning systems. Despite the improvements, the recommendation systems still required additional information for it to enhance the precision of its recommendations within a scope of applications.

The growing demand for online learning, MOOCS and smart learning systems make it a necessity for recommendation systems to provide recommendations which is both accurate and distinct to the user without any need of additional input, thereby saving time and increasing efficiency.

The system in this study introduces an online learning resource recommendation model which takes six parameters of the learner into consideration. It analyses the online behaviour and traits of the learner before making a resource recommendation. The system uses exploratory data analysis by using the Euclidean distance to identify similar resources. The similarity is analysed using association rules and

reinforced with the confidence metric. The recommendation set is prepared as the output of the rules. Finally, the prediction formula is applied to significance of a resource to a learner.

In the future enhancements for the system, it can be configured to personalise the recommendation system based on behavioural data of the user from multiple sources including social media input. It will lead to widening the subjectivity of the topics on which the resources are recommended. Given more sources for data, it can offer precise predictions.

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